

Climate Risk and Business Development Companies: Replication, Factor Maintenance, and Regulatory Capacity

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Abstract. This paper replicates CRISK for ten large U.S. banks and extends its market-based stress mechanism to business development companies (BDCs). Mean bank climate beta rises by 0.230 in 2020 (standard error 0.029), all ten banks move in the predicted direction, and replicated top-four marginal CRISK and signed CRISK increases recover 85.3 and 86.5 percent of published magnitudes. The high-frequency maximum, however, follows the November vaccine/value rotation rather than the March oil-price break. The published post-KOL fixed basket tracks KOL at 0.454 daily and 0.832 weekly; the associated BDC exposure coefficient rises from 0.025 to 0.154, but its attenuation extrapolation remains below the design's minimum detectable effect. Finally, bank $k = 8\%$ yields zero positive CRISK in every BDC-quarter. Mapping the same loss into statutory asset coverage reduces the mean buffer by 5.85 percentage points, while matched-tail climate compression is 0.343 of the market benchmark.

Keywords: climate risk; CRISK; business development companies; private credit; asset coverage

JEL Classification: G21; G23; G28; Q54

1 Introduction

The CRISK framework maps a traded stranded-asset factor into dynamic climate beta and then into a bank capital shortfall (Jung, Engle, and Berner, 2025). Extending it raises three practical questions: whether an independent archive reproduces the bank benchmark, how a post-liquidation factor should be maintained, and which capital constraint is meaningful outside banking. Business development companies (BDCs) are a useful extension because they are publicly traded private-credit intermediaries, disclose portfolio industries, and face an explicit asset-coverage requirement (Chernenko et al., 2022).

The paper reports three findings. First, mean bank climate beta rises by 0.230 in 2020 (standard error 0.029), all ten banks move in the predicted direction, and replicated top-four marginal CRISK and signed CRISK increases recover 85.3 and 86.5 percent of published magnitudes. The high-frequency path is less attributable: its November maximum follows the Pfizer–BioNTech vaccine announcement and associated value rotation, not the March oil-price break (Kapar et al., 2022).

Second, the published post-KOL continuation has a shelf life. Holding its five-stock basket fixed, KOL tracking rises from 0.454 daily to 0.832 weekly and the BDC exposure coefficient rises from 0.025 to 0.154. Yet an extrapolation to perfect tracking is 0.215, below the pooled 80-percent minimum detectable effect of 0.273. Public BDC industry disclosures therefore do not identify the exposure coefficient at this sample size.

Third, the bank capital parameter does not transfer. Applying $k = 8\%$ produces zero positive CRISK in every BDC-quarter, whereas mapping the same loss into the applicable 150- or 200-percent asset-coverage requirement reduces the mean regulatory buffer by 5.85 percentage points. At matched empirical first percentiles, climate compression is 0.343 of the market benchmark. The contribution is consequently about replication fidelity, factor maintenance, and institutional portability rather than a search for significance in every exercise.

2 Data, Factor Construction, and Method

2.1 Market, accounting, and identifier data

Daily security returns, prices, shares outstanding, and identifiers come from CRSP. Quarterly assets, equity, debt, and income come from Compustat and are linked through the CRSP/Compustat Merged table. The market archive runs from January 2010 through December 2025 and contains ten banks and twenty initially screened BDCs. The final exposure panel contains 19 BDCs with complete filings and market variables in every quarter from 2021Q1 through 2025Q4. Barings BDC (BBDC), one of the twenty market-series firms, is excluded because the maintained parser does not recover a reconciled portfolio-industry distribution for the required quarters; its returns remain in the time-series archive.

The ten-bank benchmark is Bank of America, Bank of New York Mellon, Citigroup, Goldman Sachs, JPMorgan Chase, Morgan Stanley, PNC Financial Services, State Street, U.S. Bancorp, and Wells Fargo. Bank of New York Mellon is recovered manually with CRSP PERMNO 49656 and SEC CIK 0001390777 because the restricted ticker/code extracts supplied for this archive do not contain a usable CCM/Compustat

sequence. This is an archive-specific identifier recovery, not a claim that the company is absent from Compustat. The raw link, SEC accession numbers, filing dates, and recovered quarterly fields are retained.

The balanced BDC panel contains ARCC, CGBD, CSWC, FDUS, FSK, GBDC, GSBD, HRZN, HTGC, MAIN, NMFC, OBDC, OCSL, PFLT, PSEC, SAR, SLRC, TCPC, and TSLX. Requiring all twenty quarters produces a transparent comparison set but can create survivor and disclosure-consistency selection. Results therefore describe this panel rather than the full historical BDC population.

2.2 The climate and credit factors

Following Jung, Engle, and Berner (2025), the stranded-asset factor is

$$CF_t = 0.70R_{Coal,t} + 0.30R_{XLE,t} - R_{SPY,t}. \quad (1)$$

KOL supplies the coal return through 14 December 2020. Thereafter, the coal leg follows the paper’s explicit continuation rule: a fixed weighted return of KOL’s five largest constituents immediately before liquidation. The constituents and reported final ETF weights are Washington H. Soul Pattinson (6.85 percent), Aurizon Holdings (6.19), United Tractors (6.00), China Shenhua Energy (5.97), and Adaro Energy/Alamtri Resources Indonesia (5.63). Weights are normalized within the basket. Adjusted total-return prices are converted to U.S. dollars using daily AUD/USD, IDR/USD, and HKD/USD series before log returns are formed. The SEC’s September 2020 Form N-PORT holdings provide an independent composition cross-check.

The basket is not a set of pure-play coal producers. Soul Patts is a diversified investment company; Aurizon is a rail freight and network operator; United Tractors combines equipment distribution, mining contracting, mining, construction, and energy; and China Shenhua integrates coal, power, rail, ports, and shipping.¹ More importantly for the extended sample, Adaro completes the separation of its thermal-coal assets on 5 December 2024 and the renamed Alamtri retains a 15-percent interest; the separated unit had represented 89 percent of group revenue as of June 2024 (IEEFA, 2025). Soul Patts implements its merger with Brickworks on 23 September 2025. These events do not invalidate the original short continuation, but they reduce the economic stability of a fixed basket carried through 2025.

During the common 2019–14 December 2020 overlap, the USD top-five continuation has a 0.454 daily correlation with KOL, but its weekly and monthly correlations are 0.832 and 0.931. A U.S. coal value-weighted alternative has correlations of 0.601, 0.575, and 0.783 over the same window. The 0.740 daily correlation sometimes reported for that alternative uses the longer 2010–2020 overlap and is therefore not directly comparable with the 2019–2020 top-five number. The sharp improvement for the international basket at weekly frequency is consistent with non-synchronous closes across Australia, Indonesia, Hong Kong, and the United States. Equation (1) uses SPY to match the U.S. replication design; the V-Lab

¹Company descriptions are taken from the issuers’ investor pages: <https://soulpatts.com.au/investor-centre/investor-overview>, <https://www.aurizon.com.au/who-we-are/about-us>, <https://www.unitedtractors.com/en/public-expose-2023-pt-united-tractors-tbk/>, and <https://www.csec.com/zgshwwEn/gsjj/gsjjList.shtml>.

production documentation instead identifies ACWI as its market short leg.²

The omitted-credit-factor test uses a high-yield excess-return factor rather than a short FRED spread window:

$$HY_t = \frac{1}{2}R_{HYG,t} + \frac{1}{2}R_{JNK,t} - R_{SHY,t}. \quad (2)$$

Adjusted total returns begin in 2010, so the two- and three-factor DCC models are compared over the full 2010–2025 estimation period and feed the same 380 BDC-quarter disclosure observations.

2.3 BDC investment exposure

SEC 10-Q and 10-K filings are parsed for industry distributions in the Schedule of Investments. The parser resolves split panels, retains one reconciled distribution per firm-quarter, and stores every source row, matched rule, and confidence flag. Across 10,842 extracted industry rows and 496 distinct labels, the mean mapped portfolio share is 99.76 percent. Low-confidence labels account for 6.48 percent of weight and the residual unmatched fallback for 2.43 percent.

The classification dictionary is fixed before regression estimation. Twenty-two regular-expression rules define a narrow set of energy, materials, transportation, and utility exposures. Twelve additional rules add automobiles, aerospace, capital goods, machinery, electrical equipment, construction and engineering, and related manufacturing to the broad measure. Text is normalized for capitalization, punctuation, and synonyms, so labels such as “energy,” “upstream oil and gas,” and “oil & gas” map consistently. Lower- and upper-bound tests classify every low-confidence or residual weight as non-carbon-intensive or carbon-intensive, respectively.

The broad investment share averages 10.19 percent, has a median of 10.20 percent, and ranges from zero to 38.70 percent. The narrow share averages 4.26 percent.

2.4 Asset beta and the portfolio-beta mechanism

The asset climate beta used as an outcome is a mechanical de-levering of the equity beta:

$$\beta_{i,q}^A = \beta_{i,q}^E \frac{E_{i,q}}{E_{i,q} + D_{i,q}}, \quad (3)$$

where E is market equity and D is book debt. It is not constructed from the same industry weights used on the right-hand side. Its smaller sampling dispersion follows in part from the multiplier in Equation (3). A separate mechanism variable maps each reported industry weight to an estimated Fama–French 12-industry climate beta and takes the fair-value-weighted sum. That portfolio beta is reported as a companion mechanism test, not used to construct either the equity- or asset-beta dependent variable.

²See the V-Lab CRISK methodology page at <https://vlab.stern.nyu.edu/docs/climate/CMES>.

2.5 SEC asset coverage

Asset-coverage ratios and statutory thresholds are extracted from the BDC filing archive. Actual coverage is available for 376 of 380 firm-quarters; four are unresolved. The applicable threshold is identified for every quarter: 375 observations use 150 percent and five use 200 percent. The filing value is preferred to a Compustat reconstruction because the statutory calculation includes classifications that standardized financial statements do not fully identify.

3 Empirical Method

3.1 Dynamic conditional beta

For each institution, daily returns are modeled with the market and climate factor. Conditional variances follow variance-targeted GJR-GARCH(1,1) processes and conditional correlations follow a scalar DCC(1,1) recursion. Let $F_t = (M_t, CF_t)'$. The time-varying coefficient vector is

$$\begin{bmatrix} \beta_{i,t}^M \\ \beta_{i,t}^C \end{bmatrix} = \Sigma_{FF,t}^{-1} \Sigma_{Fi,t}. \quad (4)$$

Equivalently, the climate coefficient is

$$\beta_{i,t}^C = \frac{\sigma_{i,t}}{\sigma_{c,t}} \frac{\rho_{ic,t} - \rho_{im,t}\rho_{mc,t}}{1 - \rho_{mc,t}^2}. \quad (5)$$

The model is estimated by two-step quasi-maximum likelihood. All 30 institution systems converge. Median DCC parameters are $\alpha = 0.0172$ and $\beta = 0.9794$, for persistence of 0.9966. The implied shock half-life is approximately 203 trading days. The high-yield robustness expands F_t to $(M_t, CF_t, HY_t)'$ and re-estimates the entire covariance system.

The frequency robustness compounds institution, market, and climate log returns to Friday weeks, requires at least three daily observations per week, and re-estimates the GJR-GARCH and DCC system rather than averaging the daily beta. Weekly beta is then averaged within each disclosure quarter. All 19 weekly GJR-GARCH and DCC systems converge; firm-level samples contain 337–835 weeks (median 792), and the common DCC persistence is 0.9896. This directly addresses the timing mismatch in Equation (5): asynchronous international closes can attenuate $\rho_{mc,t}$ and $\rho_{ic,t}$, so the market-removal term can be distorted rather than merely noisier.

3.2 LRMES and bank CRISK

The maintained scenario is a 50-percent decline in the climate factor over six months. Following the CRISK approximation,

$$\text{LRMES}_{i,t} = 1 - \exp[\beta_{i,t}^C \log(1 - \theta)], \quad \theta = 0.50. \quad (6)$$

The original paper interprets 50 percent as approximately the first percentile of its six-month factor

distribution. In this archive, the empirical first percentile is a 34.1-percent decline, so the maintained 50-percent scenario is more severe than the corresponding historical quantile in the 2010–2025 sample.

For banks,

$$\text{CRISK}_{i,t} = kD_{i,t} - (1 - k)(1 - \text{LRMES}_{i,t})E_{i,t}, \quad k = 0.08, \quad (7)$$

and marginal CRISK is

$$m\text{CRISK}_{i,t} = (1 - k)E_{i,t}\text{LRMES}_{i,t}. \quad (8)$$

Signed CRISK is retained for the top-four change because the published Table 2 total is the sum of signed changes. Positive-only CRISK is separately reported as a system-shortfall diagnostic.

3.3 BDC exposure regressions

The primary panel specification is

$$z(\beta_{i,q}) = \alpha + \gamma z(\text{BrownShare}_{i,q}) + \delta_q + \varepsilon_{i,q}, \quad (9)$$

where δ_q denotes calendar-quarter fixed effects. Standard errors are clustered by BDC, leaving 18 cluster degrees of freedom. Companion models use Equation (3), add standardized size, debt-to-assets, return on assets, book-to-market, and market beta, or add BDC fixed effects. Robustness tests use narrow and classification-bound exposures, lag exposure one quarter, omit 2021, and apply 4,999-repetition Rademacher wild-cluster inference.

3.4 Asset-coverage stress

Let $AC_{i,q}$ be reported asset coverage, $T_{i,q}$ the applicable threshold, $E_{i,q}$ market equity, and $D_{i,q}$ debt. The primary mapping is

$$AC_{i,q}^{\text{stress}} = AC_{i,q} - 100 \left(\frac{E_{i,q}\text{LRMES}_{i,q}}{D_{i,q}} \right). \quad (10)$$

The baseline buffer is $AC_{i,q} - T_{i,q}$ and compression is $AC_{i,q} - AC_{i,q}^{\text{stress}}$. Alternative mappings use month-end LRMES, a beta-implied monthly LRMES, positive-loss truncation, or reported book equity (NAV) in place of market equity. A 50-percent market shock uses the estimated market beta as a scale placebo. An equal-tail-probability comparison separately applies each factor’s empirical first-percentile six-month return: –34.1 percent for the climate factor and –16.8 percent for the market. Because firms have heterogeneous starting buffers, the analysis distinguishes the mean of firm-level compression-to-buffer ratios from the ratio of sample-mean compression to the sample-mean buffer.

4 Finding 1: Replication Fidelity

Table 1 and Figure 1 show that mean bank climate beta rises from 0.193 in 2019 to 0.424 in 2020. The paired increase is 0.230 (standard error 0.029, $p < 0.001$), and all ten changes are positive. The daily diagnostic gives the same point estimate with HAC(203) standard error 0.104 ($p = 0.0266$). These statistics establish a common directional movement across institutions; because all banks face the same factor realization, they are not ten independent event replications. The twenty-BDC validation is similarly positive: the mean change is 0.177 (standard error 0.023, $p = 2.48 \times 10^{-7}$), with 19 of 20 changes positive.

The magnitude comparison is also close. End-2020 top-four marginal CRISK is \$221.7 billion versus the published \$260.0 billion; the signed CRISK increase is \$372.8 billion versus \$430.89 billion. The 85.3-percent marginal-CRISK ratio equals a 0.972 equity-scale ratio times a 0.877 climate-loss-rate ratio. Panel C preserves the nonlinear identity by computing $0.92E_t\text{LRMES}_t$ daily and then averaging, rather than evaluating LRMES at mean beta.

The annual result does not isolate a single transition event. The raw cross-bank mean is 0.1974 on 9 March, 0.6329157 on 17 March, and reaches 1.0318 on 10 November after the Pfizer announcement. The 17 March value is a raw ten-bank cross-sectional mean; WFC's numerically similar 0.6328319 is a different object—its 31 December 127-day beta. The code audit verifies the two source slices and their 8.37×10^{-5} difference. Figure 2 therefore separates the oil, COVID, and vaccine/value-rotation dates rather than assigning the November maximum to the March shock.

5 Finding 2: The Continuation Rule Has a Shelf Life

Table 2 and Figure 3 summarize the continuation diagnostic. Holding the international top-five basket fixed, weekly aggregation raises KOL tracking from 0.454 to 0.832 and the equity-beta exposure coefficient from 0.025 to 0.154. This same-basket comparison isolates frequency; the U.S. coal estimate of 0.094 also changes synchronization and economic content. None rejects zero.

The deeper constraint is power. The pooled 80-percent minimum detectable effect is 0.273, while the descriptive attenuation extrapolation to perfect tracking is 0.215. Fixed-effects MDEs are 0.306–0.323 because only 15.3 percent of broad-share variation is within BDC. Thus the available 19-firm panel cannot reliably detect even the extrapolated effect. Controlled and fixed-effects signs also vary, including after high-yield adjustment. The conclusion is not that portfolio composition has no relation to climate beta; it is that public industry disclosures do not identify that relation at this sample size. Appendix B reports the full grid.

The maintenance implication is operational. A live continuation needs refresh triggers for mergers, spin-offs, and material business-mix changes. The finding applies to the documented fixed-basket rule; public documentation does not establish whether V-Lab's current production series uses that unrefreshed portfolio.

6 Finding 3: The Capital Parameter Does Not Transfer

The bank formula in Equation (7) yields zero positive CRISK for every BDC-quarter because $k = 8\%$ does not represent the BDC legal constraint. The statutory mapping in Equation (10) instead reduces mean coverage from 203.28 to 197.43 percent and the mean buffer from 53.15 to 47.30 percentage points. Table 3 reports 5.85 points of mean compression, no primary-scenario breach, and an increase from six to eleven observations within 10 points of the threshold.

The empirical first-percentile climate decline is 34.1 percent and produces 3.63 points of compression. The matched first-percentile market decline is 16.8 percent and produces 10.58 points, so the climate-to-market ratio is 0.343. The 2010–2025 window excludes 2008, making this ratio potentially high relative to a longer market-stress history. A 50-percent market scale placebo produces 33.67 points and 82 breaches. These are calibrated comparisons, not estimated structural losses; stars are uninformative because positive beta mechanically maps into positive compression and first-stage DCC uncertainty is not propagated.

7 Limitations

The BDC panel is selected on continuous survival and disclosure, portfolio labels are not borrower emissions, and only 15.3 percent of broad-share variation is within firm. The international factor is exposed to non-synchronous closes and corporate-event drift. The 2020 bank path mixes oil, pandemic, funding, and value-rotation shocks that market conditioning cannot fully separate. The coverage mapping holds debt, portfolio marks, liability classification, and management response fixed. Finally, the public V-Lab interface does not supply an unauthenticated machine-readable history; the archive provides an optional validator for an authorized series rather than reporting an unverified fit statistic.

8 Conclusion

The replication recovers the annual direction, statistical significance, and 85–87 percent of published CRISK magnitudes. Extending the factor reveals that a fixed continuation degrades with trading frequency and corporate events, while the 19-BDC exposure design lacks power to resolve the underlying coefficient. Transferring the loss mechanism also requires the institution’s actual legal constraint: bank $k = 8\%$ is vacuous for BDCs, whereas statutory asset coverage records measurable but non-breaching capacity erosion. These findings concern replication fidelity, factor maintenance, and institutional portability. Code, public outputs, and audit instructions are available at <https://github.com/MaryMai233/crisk-bdc-replication>.

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Tables and Figures

Table 1. Bank Replication of the 2020 Climate-Risk Shock.

This table evaluates the ten-bank replication through 2020. Panel A reports calendar-year mean changes. Panel B compares benchmark magnitudes and peak timing. Panel C verifies the 127-day marginal-CRISK construction. Panel D reports a separate BDC time-series validation.

	Annual beta	Daily beta	Signed CRISK	Positive CRISK
Panel A: 2020 changes				
Estimate	0.230*** (0.029)	0.230** (0.104)	343.019*** (81.640)	181.394*** (42.114)
Conventional two-sided p -value	<0.001	0.0266	<0.001	<0.001
Estimator	Paired t	HAC(203)	HAC(203)	HAC(203)
Observations	10	505	505	505
Panel B: Published-magnitude comparison				
	Replicated	Published	Ratio	Qualification
Annual-average peak year	2021	2020	—	2020–2021 plateau
Raw daily peak	1.0318	—	—	Vaccine/value rotation
Maximum 127-day beta (WFC)	0.6328	< 0.500	1.27	Smoothing not stated
Top-four m CRISK (USD bn)	221.7	260.0	0.85	85.3%
Top-four CRISK increase (USD bn)	372.8	430.9	0.87	86.5%
Panel C: End-2020 top-four 127-day mean construction				
Institution	Mean beta (reference)	Mean $E \times \text{LRMES}$	Mean m CRISK	Identity error
BAC	0.5815	73.32	67.5	0.00e+00
C	0.6326	37.18	34.2	0.00e+00
JPM	0.5015	93.69	86.2	0.00e+00
WFC	0.6328	36.76	33.8	7.28e-15
Total	—	240.94	221.7	7.28e-15
Panel D: BDC time-series validation				
Sample	Mean beta 2019	Mean beta 2020	Mean change (SE)	Paired p
20 BDCs	0.068	0.245	0.177 (0.023)	2.48×10^{-7}
19-BDC exposure sample	0.065	0.244	0.179 (0.024)	5.84×10^{-7}

Notes: Panel A's CRISK estimates are differences between daily calendar-year means. Daily diagnostics use HAC(203); lag sensitivity at 21, 63, 126, and 203 days is archived. Because institutions share one factor realization, paired statistics measure cross-institution consistency rather than independent event replications. Panel B's benchmark-aligned change uses December-average beta with year-end debt and market equity. Panel C first applies $m\text{CRISK} = 0.92E \times \text{LRMES}$ daily and then averages; mean beta is reference only. The 0.853 $m\text{CRISK}$ ratio equals 0.972 times 0.877. Panel D is a time-series validation, not the exposure regression. ***, **, and * denote significance at 1, 5, and 10 percent.

Table 2. Factor Continuation, Tracking, and Detectable Effects.

This table compares the published top-five continuation at daily and weekly frequencies with a U.S. coal alternative. Coefficients relate standardized broad investment share to standardized BDC climate beta with quarter fixed effects and BDC-clustered standard errors.

	Top-five daily		U.S. coal daily		Top-five weekly	
	Equity	Asset	Equity	Asset	Equity	Asset
Broad investment share	0.025 (0.093)	0.014 (0.089)	0.094 (0.102)	0.114 (0.078)	0.154 (0.238)	0.149 (0.185)
Two-sided p -value	0.794	0.875	0.368	0.164	0.524	0.430
KOL tracking correlation	0.454	0.454	0.601	0.601	0.832	0.832
Observations	380	380	380	380	380	380
Pooled MDE_{80}	0.273	0.260	–	–	–	–
Correlation-one extrapolation	0.215	–	–	–	–	–

Notes: Weekly DCC is re-estimated from compounded Friday-week returns; all 19 systems converge. The daily-to-weekly top-five comparison holds economic content fixed and changes frequency. The U.S. coal portfolio changes both synchronization and content. The extrapolation is descriptive, uses three construction-frequency points, and is not an errors-in-variables correction. No coefficient is statistically significant.

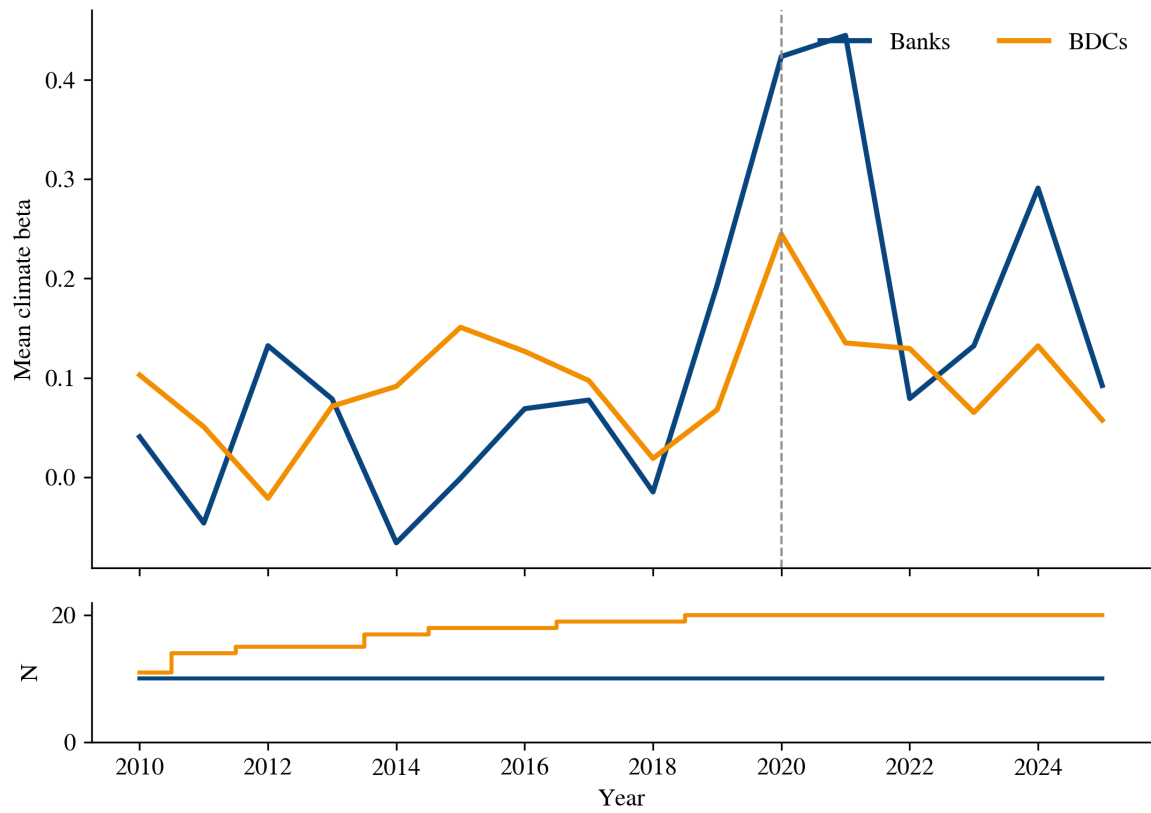
Table 3. Climate Stress and BDC Asset-Coverage Capacity.

Panel A compares the maintained climate shock, empirical-tail climate and market shocks, a market scale placebo, and NAV mapping. Panel B reports LRMES-timing sensitivity. Panel C gives the mechanical portfolio-exposure link.

Panel A: Scenario comparison					
	Climate 50%	Climate <i>p</i> 01	Market <i>p</i> 01	Market 50%	Climate/NAV
Mean buffer compression (pp)	5.849	3.633	10.583	33.671	5.908
Median buffer compression (pp)	5.734	3.511	9.367	29.860	5.601
Mean compression / mean buffer (%)	11.01	6.84	19.91	63.35	11.11
Legal breaches	0	0	0	82	0
Observations	376	376	376	376	376
Panel B: Climate 50-percent timing and loss mapping					
	Monthly mean	Month-end	Beta-implied	Positive-loss	
Mean buffer compression (pp)	5.849	6.449	5.899	6.515	
Median buffer compression (pp)	5.734	5.665	5.751	5.734	
Within 10 pp after stress	11	15	11	13	
Legal breaches	0	1	0	0	
Panel C: Mechanical exposure-to-capacity link					
Exposure coefficient			0.0245		
Predicted beta change, zero to maximum share			0.0128		
Representative incremental compression			0.704 pp		
Mean increment at observed exposures			0.169 pp		

Notes: The *p*01 columns apply each factor's separate marginal first percentile: −34.1 percent for climate and −16.8 percent for the market. They are not a joint tail state. The market window excludes 2008. The 50-percent market shock is a scale placebo. NAV replaces market equity. Panel C is a deterministic transformation of an imprecise exposure coefficient, not a causal estimate. No significance markers are reported because positive beta mechanically maps into positive compression and first-stage DCC uncertainty is omitted.

Figure 1. Annual Climate Beta for Banks and BDCs.



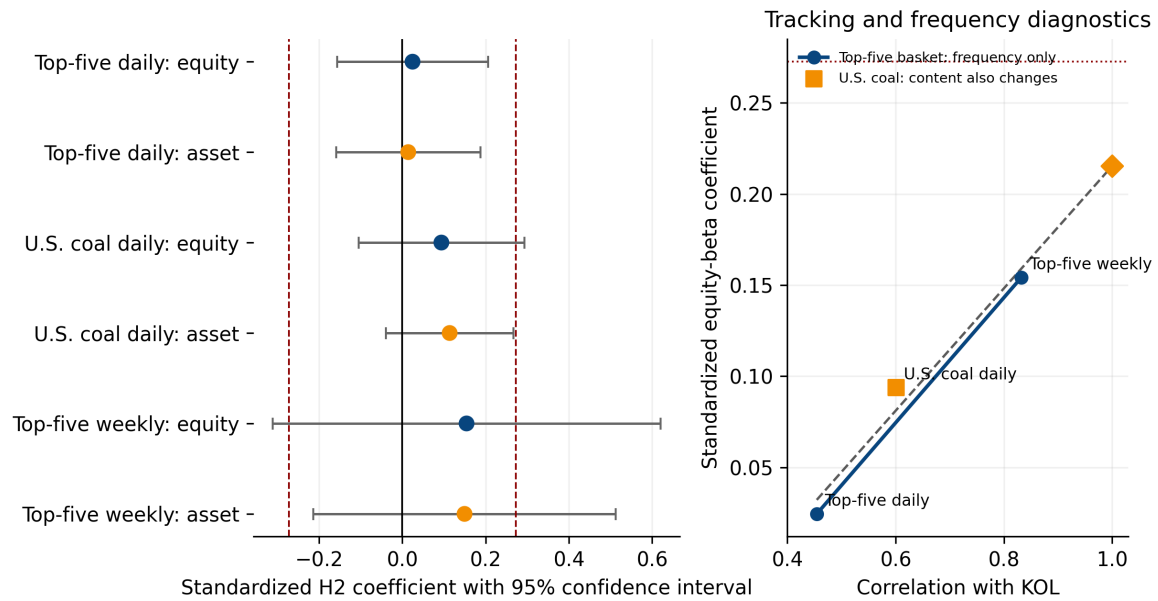
Notes: The upper panel plots annual cross-sectional mean two-factor dynamic conditional climate beta; the lower panel reports the number of active institutions. The vertical line marks 2020. KOL is used through 14 December 2020 and the fixed USD top-five constituent return thereafter. Changes in the BDC series partly reflect entry and exit from the active sample.

Figure 2. Top-Four Bank CRISK Around the 2020 Shock.



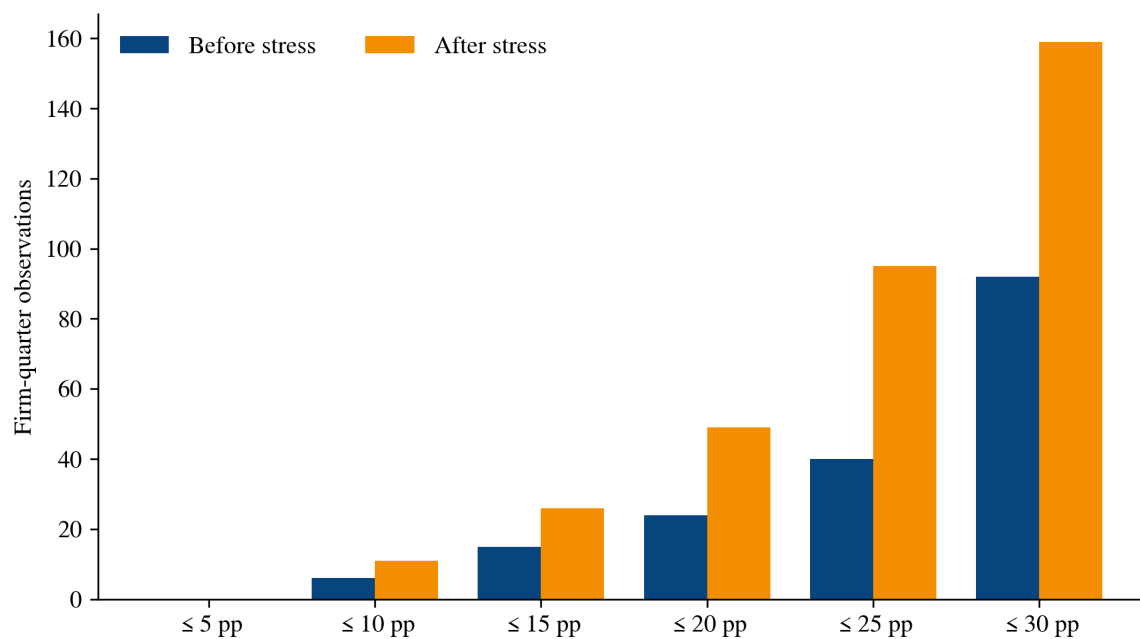
Notes: The upper panel reports raw daily series; the lower panel reports trailing 127-trading-day means for BAC, C, JPM, and WFC. Signed CRISK retains capital surpluses; positive CRISK truncates negative institution values at zero. Vertical markers identify the 9 March oil-price break, the 16 March COVID market dislocation, and the 9 November Pfizer–BioNTech announcement. Smoothing is shown separately so it cannot obscure event timing.

Figure 3. Exposure Sensitivity to Factor Construction.



Notes: The left panel reports standardized broad-investment-share coefficients and 95-percent confidence intervals; dashed vertical lines mark the pooled 80-percent MDE of ± 0.273 . In the right panel, the connected blue top-five points hold basket membership fixed and vary only return frequency. The orange U.S. coal point changes both synchronization and economic content. The dashed line and correlation-one diamond are a descriptive three-point extrapolation, not a formal errors-in-variables correction; no fit statistic is interpreted with $n = 3$. None of the estimates is statistically different from zero at conventional two-sided levels.

Figure 4. Proximity to the Statutory Asset-Coverage Threshold.



Notes: Bars count BDC-quarter observations within the indicated number of percentage points of the applicable 150- or 200-percent threshold. Categories are cumulative. No primary-scenario observation crosses the threshold.

A BDC Descriptive Statistics

Table A1. BDC Panel Descriptive Statistics.

Variable	<i>N</i>	Mean	SD	Median	Minimum	Maximum
Equity climate beta	380	0.098	0.104	0.094	-0.276	0.409
Asset climate beta	380	0.047	0.049	0.045	-0.103	0.194
Broad carbon-intensive share (%)	380	10.190	7.681	10.200	0.000	38.700
Narrow carbon-intensive share (%)	380	4.258	2.843	4.300	0.000	11.834
Total assets (USD mn)	380	5,257.993	6,055.510	3,081.480	454.118	31,235.000
Debt to assets	380	0.507	0.071	0.517	0.283	0.689
Quarterly ROA	380	0.011	0.012	0.011	-0.072	0.056
Book to market	380	1.0278	0.2500	1.0312	0.5026	2.6256
Market beta	380	0.642	0.215	0.629	0.205	1.354
Reported asset coverage (%)	376	203.283	39.438	187.650	155.700	347.100
Market to NAV	380	1.0277	0.2496	0.9697	0.3809	1.9897

Notes: Asset beta mechanically de-levers equity beta. Book-to-market and market-to-NAV are observation-level reciprocals; the maximum product-minus-one error is 5.6×10^{-16} . Within-BDC movements account for 15.3 percent of broad-share variance. Four observations lack a resolved coverage ratio.

B Exposure Power and Robustness

Table B1. BDC Investment Exposure: Full Specification Grid.

Panel A: Main specifications							
	(1)	(2)	(3)	(4)	(5)	(6)	
Broad investment share	0.025 (0.093)	0.014 (0.089)	-0.018 (0.052)	-0.052 (0.059)	-0.062 (0.104)	-0.099 (0.110)	
Outcome	Equity	Asset	Equity	Asset	Equity	Asset	
Financial controls	No	No	Yes	Yes	No	No	
BDC fixed effects	No	No	No	No	Yes	Yes	
Quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	
Observations	380	380	380	380	380	380	
Panel B: Classification, timing, and inference							
	Narrow	Lower	Upper	Lagged	Post-2021	Wild p	MDE ₈₀
Equity beta	-0.007 (0.076)	0.021 (0.092)	-0.002 (0.067)	0.029 (0.097)	-0.004 (0.103)	0.780	0.273
Asset beta	-0.021 (0.071)	0.015 (0.087)	-0.005 (0.073)	0.012 (0.089)	-0.016 (0.097)	0.863	0.260
Observations	380	380	380	361	304	380	380
Panel C: High-yield excess-return factor							
	Baseline eq.	HY equity	Baseline asset	HY asset	HY + controls	HY + BDC FE	
Broad investment share	0.025 (0.093)	0.078 (0.134)	0.014 (0.089)	0.048 (0.117)	-0.007 (0.076)	0.102 (0.140)	
Two-sided p -value	0.794	0.569	0.875	0.688	0.928	0.475	
Observations	380	380	380	380	380	380	

Notes: Outcomes and exposures are standardized. All models include quarter fixed effects and cluster standard errors by BDC unless indicated. Financial controls are size, debt-to-assets, ROA, book-to-market, and market beta. Lower and upper bounds reclassify uncertain weight; lagged exposure uses the prior quarter; Post-2021 omits 2021. Wild p uses 4,999 Rademacher replications. No coefficient reaches conventional two-sided significance.